

4-bit adapters can match 16-bit accuracy on standard benchmarks

Preserving 16-bit performance while finetuning 4-bit models

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Finetune 65B on a single 48GB GPU

4-bit NF4 + Double Quantization

Paged Optimizers

Guanaco family: up to 99.3% of ChatGPT on Vicuna

Code: github.com/artidoro/qlora • github.com/TimDettmers/bitsandbytes

Memory/Perf Innovations

Scale

Backprop through 4-bit LLM into LoRA matches 16-bit baselines

- Backprop through frozen 4-bit quantized LLM into LoRA adapters.
- Matches 16-bit finetuning performance; massive memory savings.
- Guanaco models: state-of-the-art open chatbots on Vicuna.

Summary of abstract

4-bit finetuning fits in <48 GB without speed loss

- 16-bit finetuning of 65B needs >780 GB GPU memory.
- Prior quantization great for inference, breaks in training.
- QLORA enables 4-bit finetuning with no performance loss.
- 65B finetuning fits in <48 GB; runtime preserved.

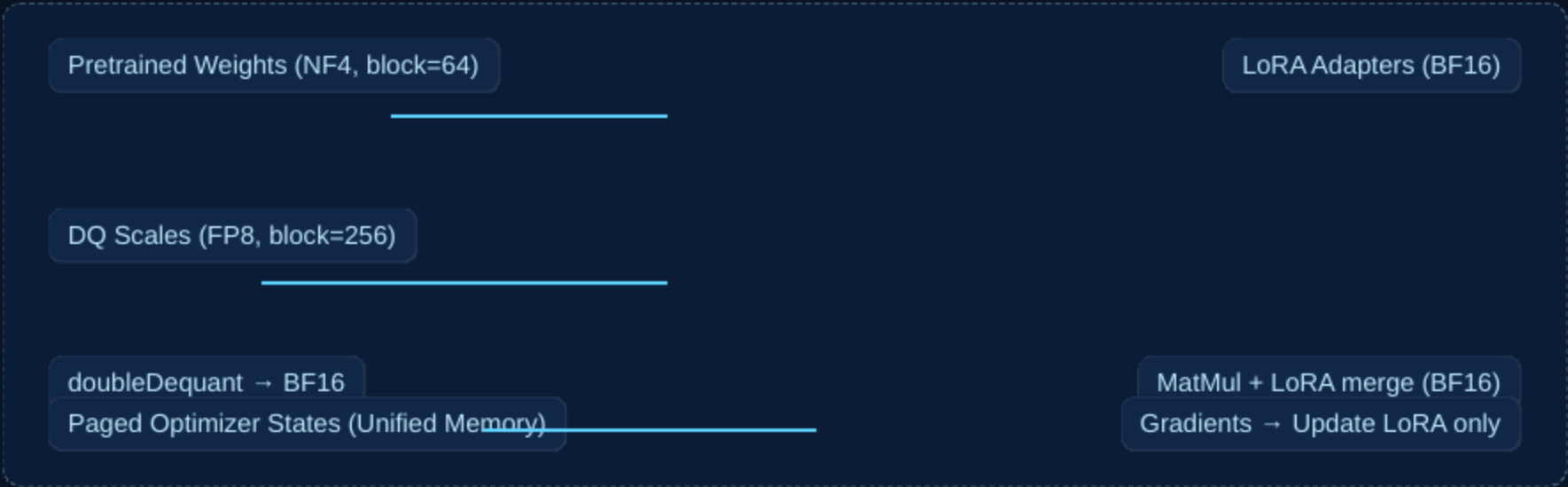
Figure placeholder — Memory requirements of finetuning methods (QLORA vs LoRA vs full FT)

Figure 1: QLORA quantizes base model to 4-bit and uses paged optimizers

Block-wise NF4 + LoRA enables accurate 4-bit training

QLORA: 4-bit storage → BF16 compute via LoRA; paged optimizers prevent OOM

- Storage: NF4 (4-bit); Compute: BF16; Update: adapters only.



NF4 + DQ

- NormalFloat bins incl. zero; per-block scales.
- Quantized scales cut overhead to ~0.127 b/param.

Paged Optimizers

- Unified memory pages optimizer state to CPU.
- Avoids checkpointing memory spikes.

LoRA on all transformer linear layers matches 16-bit finetuning

- Placement/coverage drives parity; r is less critical.
- Strong 16-bit baselines require tuned LR and batch sizes.

Our 5-shot and GPT-4 judging produce consistent, fair comparisons

- Standardized 5-shot MMLU and tournament-style Vicuna/OA pairwise judging.
- Human and GPT-4 raters yield aligned system-level rankings (moderate agreement).
- Elo with 10k orderings (K=32) mitigates ordering and scale biases.

Section 5.2

Takeaway: QLORA retains 16-bit adapter accuracy while slashing memory, enabling 33B/65B finetuning on single GPUs

QLORA vs 16-bit LoRA: same accuracy, fraction of memory

Setup

- Instruction-tuned LLaMA (7B–65B) on Alpaca/FLAN v2.
- Metrics: 5-shot MMLU, Vicuna (GPT-4), Memory.

Key Metrics

- MMLU: NF4+DQ $\approx$ BF16 LoRA across sizes (Table 4).
- Vicuna: Guanaco-65B reaches 99.3% of ChatGPT (Table 6).
- Memory: 65B fits in <48 GB vs >780 GB for full 16-bit FT.

4-bit QLORA matches 16-bit accuracy and cuts memory to <48 GB

NF4+DQ achieves 16-bit LoRA/full-FT parity while enabling single-GPU finetuning.



- MMLU: NF4+DQ ≈ BF16 LoRA across 7B–65B (Table 4).
- Vicuna: Guanaco-65B 99.3%, 33B 97.8%, 7B 87.0 (Table 6).
- Memory: 65B finetuning on a single 48GB GPU.

Main results

Caveats: moderate human ↔ GPT-4 agreement; GPT-4 order/self-preference effects noted.

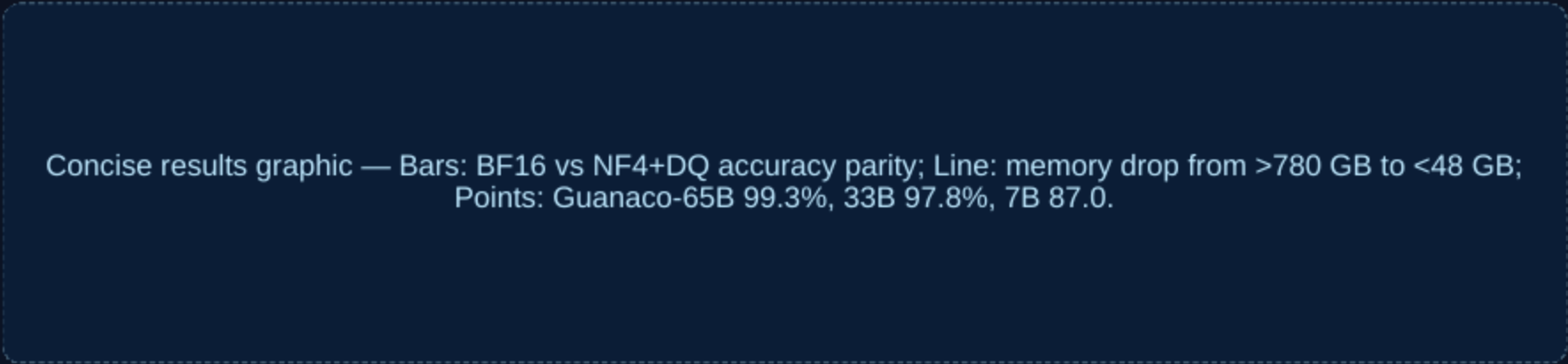
Strengths	Caveats
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Evaluation validity: consistent rankings, but known biases remain

Section 6 — Validity

4-bit training matches 16-bit while slashing memory

Backup



Empirical message

- NF4+DQ QLORA ≈ 16-bit LoRA/full FT on standard benchmarks.
- Guanaco tops open models; near ChatGPT on Vicuna.

Efficiency

- 65B finetuning on a single 48GB GPU.
- Paged optimizers avoid OOM on long sequences.

Backup

Releasing code and adapters enables full reproduction

- Code and kernels: <https://github.com/artidoro/qora> • <https://github.com/TimDettmers/bitsandbytes>
- Models: Adapters for LLaMA 7/13/33/65B across 8 datasets (32 models).
- Evaluation assets: model generations with human and GPT-4 annotations